



Matrine protects against experimental autoimmune encephalomyelitis through modulating microglial ferroptosis

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ABSTRACT

Multiple sclerosis (MS) is an inflammatory demyelination neurodegenerative disease of the central nervous system (CNS). Ferroptosis has been implicated in a range of brain disorders, and iron-loaded microglia are frequently found in affected brain regions. However, the molecular mechanisms linking ferroptosis with MS have not been well-defined. The present study seeks to bridge this gap and investigate the impact of matrine (MAT), a herbal medicine with immunomodulatory capacities, on the regulation of oxidative stress and ferroptosis in the CNS of mice with experimental autoimmune encephalomyelitis (EAE), an animal model of MS. CNS of EAE mice contained elevated levels of ferroptosis-related molecules, e.g., MDA, LPCAT3 and PTGS2, but decreased expression of antioxidant molecules, including GSH and SOD, GPX4 and SLC7A11. This pathogenic process was reversed by MAT treatment, together with significant reduction of disease severity and CNS inflammatory demyelination. Furthermore, the expression of PTGS2 and LOX was largely increased in microglia of EAE mice, accompanied with increased production of IL-6 and TNF- α , indicating a proinflammatory phenotype of microglia that undergo oxidative stress/ferroptosis, and their expression was significantly reduced after MAT treatment. Together, our results indicate that ferroptosis/inflammation plays an important role in the pathogenesis of CNS autoimmunity, and inhibiting ferroptosis-induced microglial activation/inflammation could be a novel mechanism underlying the therapeutic effects of MAT on CNS inflammatory demyelination in EAE.

1. Introduction

Multiple sclerosis (MS) is an inflammatory demyelinating disorder in the central nervous system (CNS), which is more common in young and middle-aged individuals [1]. The existing therapeutic approaches for MS, mainly by modulating the immune response [2,3], are largely constrained because of the complicated and diverse genesis and progression of this disease [4–6]. However, the control of other relevant components, such as lipid peroxidation, has long been neglected in the

long-term treatment of neuroinflammation.

Ferroptosis, a form of programmed cell death mediated by iron-dependent peroxidation, may present a potential mechanism underlying the pathogenesis of MS [7–10] and its animal model experimental autoimmune encephalomyelitis (EAE) [11]. This distinct mode of programmed cell death, distinguishing it from apoptosis and necrosis, may result from the increased levels of oxidative stress (OS) in MS [12]. During cell ferroptosis, the activation of immune cells triggers inflammation of the targeted tissues, concomitant with the upregulation of

Abbreviations: MS, multiple sclerosis; CNS, central nervous system; OS, oxidative stress; MAT, matrine; EAE, experimental autoimmune encephalomyelitis; GPX4, glutathione peroxidase 4; SLC7A11, solute carrier family 7 member 11; LPCAT3, lysophosphatidylcholine acyltransferase 3; PTGS2, prostaglandin-endoperoxide synthase 2; ROS, reactive oxygen species; Elisa, enzyme linked immunosorbent assay; NDs, neurodegenerative diseases; GSH, glutathione; LOX, Lipoxigenase; PTGS, prostaglandin-endoperoxide synthase; MOG, myelin oligodendrocyte glycoprotein; PBS, phosphate buffered solution; HE, hematoxylin-eosin; LFB, luxol fast blue; PMSF, phenylmethanesulfonyl fluoride; BCA, Bicinchoninic Acid Assay; PVDF, poly vinylidene fluoride; HRP, horseradish peroxidase; IL-1 β , interleukin-1 β ; TNF- α , tumor necrosis factor-alpha; IL-6, interleukin-6; IBA-1, ionized calcium binding adapter molecule 1.

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specific enzymes that promote the production of additional inflammatory mediators [13–15]. This process could be restrained by antioxidant mechanisms; for example, SLC7A11, a member of the solute carrier family seven, acts as the catalytic subunit of the xCT system, regulating the cellular iron death defense system and generating glutathione (GSH) [16–18]. Glutathione, together with intracellular glutathione peroxidase 4 (GPX4), forms a crucial antioxidant pathway [18–20]. Defective expression of these modulatory molecules leads to the accumulation of OS/ferroptosis factors in targeted tissues, and restoring intracellular glutathione levels to repair the antioxidant mechanism is, therefore, a potential approach to combating OS/ferroptosis-induced tissue damage.

Matrine (MAT), an alkaloid derived from traditional herbal medicine, has been found to regulate immune cell activities and reduce inflammatory factors, thus maintaining a favorable cellular microenvironment [21,22]. MAT has been shown to be effective in suppressing EAE [23], while the mechanism of its action has not been fully understood. MS shares common hallmarks with other neurodegenerative diseases, such as inflammation and OS [24], and recent studies have suggested a potential link between MS/EAE and ferroptosis [8,9,11]; however, if MAT impacts this process is unknown. The present study aims to investigate the mechanism of MAT action and its potential therapeutic application in MS by modulating the ferroptosis pathway and regulating microglial cell-mediated iron death processes. This research may have potential to offer a novel approach to regulating OS/ferroptosis pathway in MS/EAE.

2. Materials and methods

2.1. Antibodies and reagents

Anti-SLC7A11 (Proteintech Group, Wuhan, China), anti-GPX4 (Proteintech Group, Wuhan, China), anti-PTGS2 (Proteintech Group, Wuhan, China), anti-LPCAT3 (Proteintech Group, Wuhan, China) and anti-GAPDH (Proteintech Group, Wuhan, China), anti-iba1 Mouse Polyclonal Antibody (Servicebio, Wuhan, China), anti-IL-6 Rabbit Polyclonal Antibody (Servicebio, Wuhan, China), PTGS2 antibody Mouse (Proteintech Group, Wuhan, China), anti-TNF- α Rabbit Polyclonal Antibody (Servicebio, Wuhan, China), LOX antibody Rabbit (Abmart, Shanghai, China), antibodies-goat anti-rabbit HRP (Servicebio, Wuhan, China), antibodies-goat anti-rabbit Cy3 conjugate (Proteintech Group, Wuhan, China), goat anti-mouse Alexa Fluor 488 (Servicebio, Wuhan, China), Heat-killed Mycobacterium tuberculosis powder (BD, Batch number 231141, USA), Freund's incomplete adjuvant (Sigma, St. Louis, MO, USA), MOG35-55 peptide powder (Kingsley Biotechnology Co., Nanjing, China), pertussis toxin (Sigma-Aldrich, Brøndby, Denmark), MAT (Shanghai yuan ye Bio-Technology Co, Ltd, China), tribromoethanol (Nanjing Ai bei Biotechnology Co., Ltd, China), BCA Protein Concentration Assay Kit (Servicebio, Wuhan, China), transcription kit (TaKaRa, Japan).

2.2. Animals

Thirty female C57BL/6 mice, 8–10 weeks old and weighing 20 ± 2 g, were supplied by the Laboratory Animal Center of Henan Provincial. All mice were maintained in the SPF grade animal house of Zhengzhou University Laboratory Animal Center. Mice were group housed (up to 5 per cage), food and water were provided with aseptic levels by animal platforms. All animal experiments were approved by the Ethics Committee of Zhengzhou University and our reporting follows the recommendations in the ARRIVE guidelines. All methods were carried out in accordance with relevant guidelines and regulations.

2.3. Induction and scoring of EAE

The experimental animal modeling method for EAE was established with reference to the Bittner et al. [25]. Heat-killed Mycobacterium

tuberculosis powder (Batch number 231141 purchased from BD Company in the United States) was diluted to 4 mg/ml with Freund's incomplete adjuvant (Sigma, St. Louis, MO, USA) to prepare the homogeneous suspension A. Stock solution was prepared by dissolving the dried MOG35-55 peptide powder (Kingsley Biotechnology Co., Nanjing, China) using sterile water. Dilute the stock solution with anhydrous PBS to obtain the B solution (2 mg/ml). Mixing liquid A and B. Dilute pertussis toxin (PTX, Sigma-Aldrich, Brøndby, Denmark) and prepare a working solution of 2 ng/ μ l. Mice were subcutaneously (s.c.) injected with 200 μ l of antigen emulsion at two symmetrical locations on the back and at the nape of the neck, followed by 200 ng of PTX on day zero and day two post immunization (p.i.) [26]. To minimize the impact of processing sequence as much as possible, we will increase the number of experimental operators to minimize processing time.

To further investigate the effect of MAT on the mouse model of multiple sclerosis, MAT was administered intraperitoneally according to a previous protocol [27]. The immunized mice were randomly divided into two groups ($n = 10$ each group): (1) EAE + Saline: immunized mice received an equal volume of Saline per day from days 0–25 p.i. (2) EAE + MAT: Previous research has shown that 50 mg/kg/day is the optimal dose for animal administration [27]. Immunized mice were treated with MAT (50 mg/kg/day, Shanghai yuan ye Bio-Technology Co, Ltd, catalog: 16837-52-8) [27–29]. Synchronously establishing a blank control group ($n = 10$).

Clinical signs of EAE were scored daily by the double-blind manner as follow: 0 = normal agility of movement, mentally active, 1 = Flaccid tail, unable to curl and curl up, 2 = Weakness in the hind limbs and uncoordinated movement, 3 = paralyzed hind limbs, unable to stand upright, 4 = Quadriplegic, unable to move, 5 = moribund slow breathing or dead.

2.4. Histopathological evaluation

For pathological experiments, mice were subjected to deep anesthesia with tribromoethanol (0.2 ml/10 g, Nanjing Ai bei Biotechnology Co., Ltd, catalog: MA0478). Tissue samples were collected from mice by cardiac perfusion on day 25 p.i. The spinal cord at the lumbar expansion was individually intercepted and immersed in 4 % paraformaldehyde for 48 h, while the rest of the fresh spinal cord was snap-frozen in liquid nitrogen, and then transferred to a -80°C refrigerator for spare parts. Paraformaldehyde-fixed tissue was embedded in paraffin for tissue sectioning, and 5- μ m-thick sections were stained for pathologic evaluation. After the process of making water, Hematoxylin-eosin (HE) staining and Luxol Fast Blue (LFB) staining were conducted to detect inflammatory infiltration and demyelination. Inflammatory infiltrates were scored on the following scale [11]: 0 = no inflammatory cell infiltration; 1 = Few inflammatory cells infiltrated; 2 = Presence of inflammatory cell infiltration around blood vessels; 3 = moderate inflammatory cell infiltration in the parenchyma; 4 = Severe cellular infiltration in the parenchyma. The scoring criteria for spinal cord demyelination are as follows: 0 = no demyelination symptoms; 1 = Localized myelin loss; 2 = Presence of areas of demyelination; 3 = Massive fused myelin loss; 4 = extensive demyelination. Three tissue sections were dissected for each sample, the pathological state was analyzed according to the evaluation criteria and the mean value was calculated.

2.5. Western blot

Tissue frozen at -80°C was taken and weighed, and 10 μ l of protein extract was added per 1 mg (Radioimmunoprecipitation Lysis Buffer) lysis fluid: PMSF (phenylmethylsulfonyl fluoride) protease inhibitor = 100:1 proportional allocation. Protein concentration was measured using the BCA Protein Concentration Assay Kit, and a standard curve was plotted to calculate the protein electrophoretic uptake. Protein-attached PVDF membranes were obtained through gel preparation,

Table 1
Sequences of the primer pairs.

Target cDNA	Primer sequence (5' to 3')
GAPDH	GAPDH-F: GGTGTCTCTCCTGGACTTCA GAPDH-R: TGGTCCAGGGTTTCTTACTCC
GPX4	GPX4-F: CCTCCCCAGTACTGCAACAG GPX4-R: GCACACGAAACCCCTGTACT
SLC7A11	SLC7A11-F: GTGAGATGGGCCAGCAGAAAG SLC7A11-R: CAGCCTCCGCTCATATTGC
PTGS2	PTGS2-F: TGAGTACCGCAAACGCTTCT PTGS2-R: CAGCCATTCTCTCTCTCTGT
LPCAT3	LPCAT3-F: ACTGAAGCTAATTGGGCTGTGT LPCAT3-R: TCCAGCAATGAAGGGACACC

electrophoresis, and membrane transfer, closed in 5 % skim milk for 1 h, wash with 0.05 % PBS at room temperature for 5 min $\times$ 3 times, then incubated overnight at 4 °C with rabbit anti-SLC7A11 (Proteintech Group, Wuhan, China), rabbit anti GPX4 (Proteintech Group, Wuhan, China), rabbit anti-PTGS2 (Proteintech Group, Wuhan, China), mouse anti-LPCAT3 (Proteintech Group, Wuhan, China) and rabbit anti-GAPDH (Proteintech Group, Wuhan, China). The PVDF membrane was incubated with the corresponding HRP secondary antibody for 1 h at room temperature and washed for 5 min at room temperature using PBS containing 0.05 % PBS $\times$ 3 times. The strips were then processed by exposure using an ultrasensitive chemiluminescence kit. The relative protein expression was inferred by calculating the gray value of the target protein on the strip using IMAGE J software.

2.6. Immunofluorescence

The paraffin-embedded tissues were sliced and fixed on slides. After melting the paraffin and fixing the tissues, the slides underwent elution with xylene and anhydrous ethanol, followed by immersion in graded alcohols and tissue repair. Immunofluorescence blocking solution was applied, followed by overnight incubation with the primary antibody at

4 °C and subsequent incubation with the secondary antibody at room temperature. Finally, the slides were mounted and observed under a fluorescence microscope [30]. Primary antibodies included anti-iba1 Mouse Polyclonal Antibody (Servicebio, Wuhan, China), anti-IL-6 Rabbit Polyclonal Antibody (Servicebio, Wuhan, China), PTGS2 antibody Mouse (Proteintech Group, Wuhan, China), anti-TNF-alpha Rabbit Polyclonal Antibody (Servicebio, Wuhan, China) and LOX antibody Rabbit (Abmart, Shanghai, China). Secondary antibodies contained antibodies-goat anti-rabbit HRP (Servicebio, Wuhan, China), antibodies-goat anti-rabbit Cy3 conjugate (Proteintech Group, Wuhan, China), goat anti-mouse Alexa Fluor 488 (Servicebio, Wuhan, China). All images were captured by confocal microscope (Olympus Fluoview FV1000). Six sections from each group were examined in a blind manner. Co-expression of target proteins with target cells was calculated using IMAGE J software.

2.7. Quantitative real-time polymerase chain reaction (qRT-PCR)

Total RNA was extracted from spinal cord tissue using RNAiso plus. RNA concentration was determined using a Nano Drop ONE spectrophotometer (Thermo, USA). The cDNA was diluted to the appropriate concentration according to the concentration ratio, after which the cDNA was reverse transcribed using a reverse transcription kit (TaKaRa, Japan). For quantitative analysis of cDNA, cDNA levels were determined using a Nano Drop ONE spectrophotometer (Thermo, USA), afterwards, the upstream and downstream primers of the target gene and the fluorescent dye are added (Hieff® qPCR SYBR Green Master Mix No Rox, Yeasen Biotechnology). The instrument detects the expression of the target gene, and all operation procedures are conducted in full accordance with the product manual. GAPDH as a reference for target genes. The relative expression levels of target genes were calculated using the $2^{-\Delta\Delta T}$ method. The sequences of the primers are shown in Table 1.

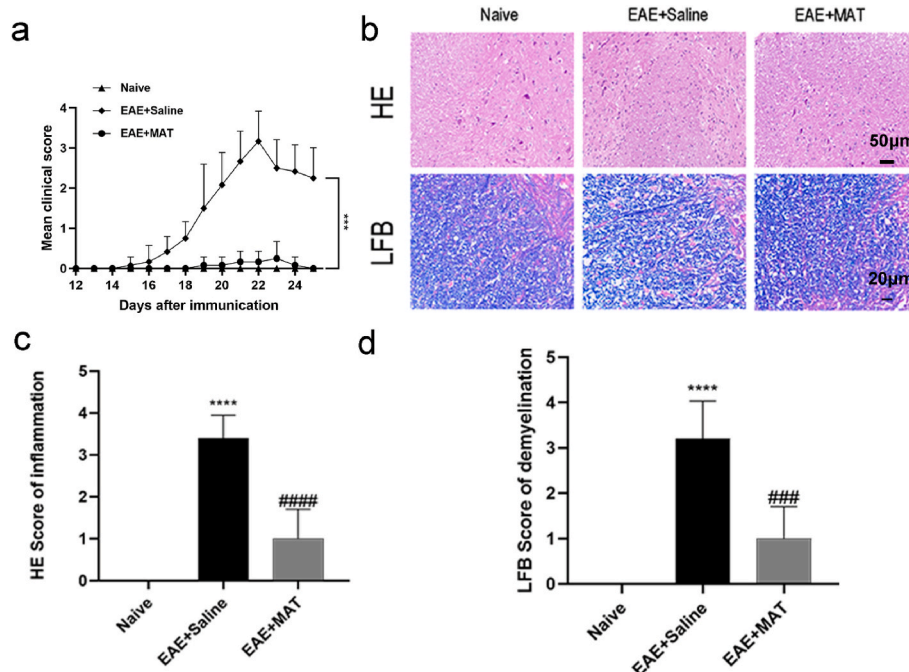


Fig. 1. MAT reduced disease progression and suppressed immune cell infiltration in EAE. Female C57Bl/6 mice were injected s.c. with MOG35-55 peptide in CFA to induce EAE. From day 12 p.i. onwards, mice were intraperitoneally (i.p.) injected daily with physiological saline (EAE + Saline) or MAT (EAE + MAT). (a) Clinical severity was blindly scored daily by two researchers (n = 5 per group). (b) Mice were sacrificed on day 25 p.i. Brain and spinal cord were harvested after extensive perfusion, and H&E and LFB staining in spinal cord were performed. Spinal cords of naive mice served as control. (c, d) Data from b are shown as mean $\pm$ SD (n = 5 per group). * P < 0.05, *** P < 0.001, **** P < 0.0001, compared to naive mice, ### P < 0.001, #### P < 0.0001, compared to saline-treated EAE mice.

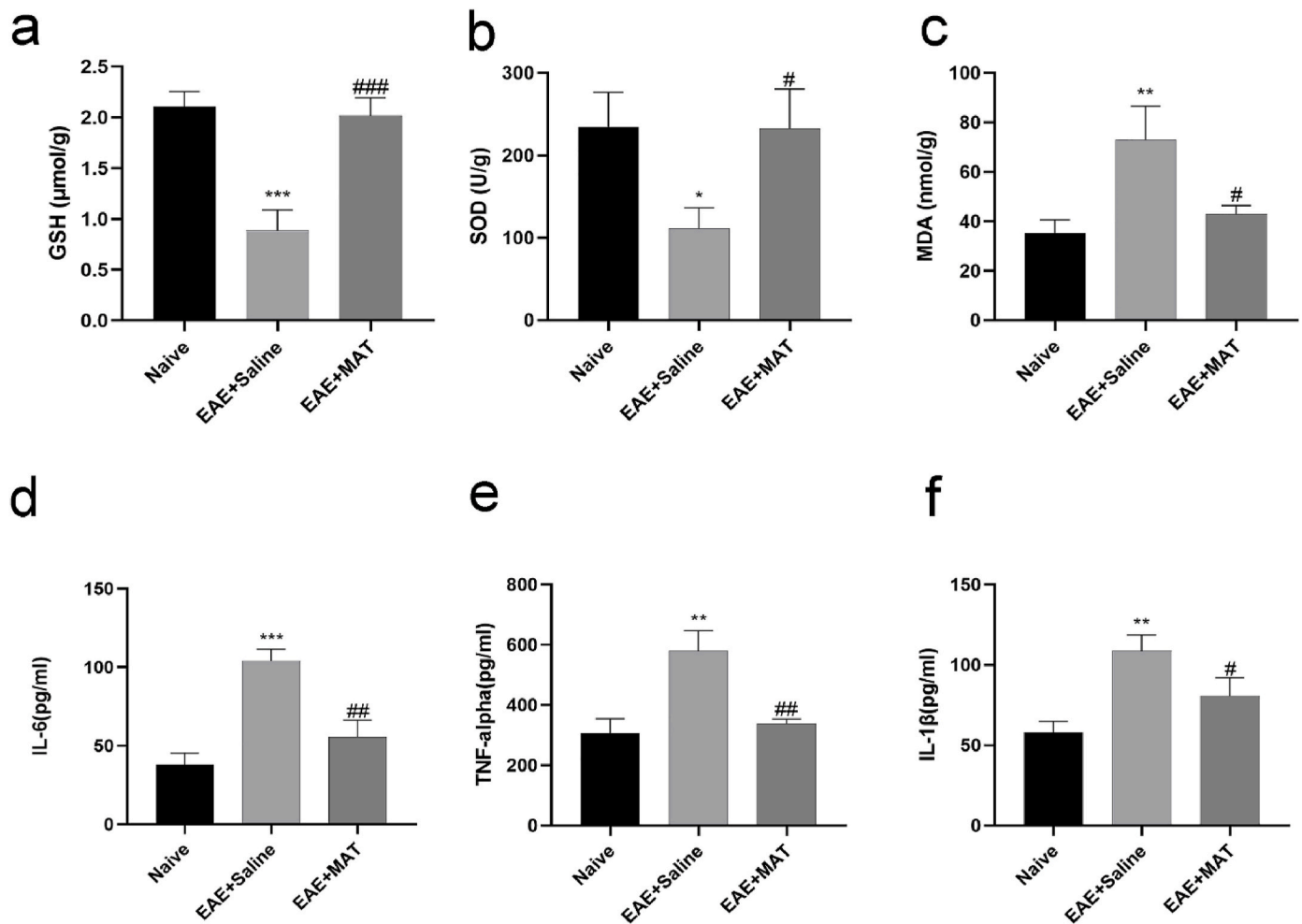


Fig. 2. MAT reduced levels of oxidative stress and proinflammatory factors in EAE mice. At day 25 p.i., brains from each individual mouse were harvested following extensive perfusion, and the concentrations of GSH, SOD and MDA (a–c) and IL-6, TNF- α , and IL-1 β (d–f) in the brain homogenate were measured by ELISA. Data represent mean $\pm$ SD (n = 4 per group). * P < 0.05, ** P < 0.01, *** P < 0.001, compared to naive mice; # P < 0.05, ## P < 0.01, ### P < 0.001, compared to EAE + Saline mice.

2.8. ELISA

Brain tissues of animals were collected after 25 days of execution and quickly preserved in liquid nitrogen, after which appropriately sized blocks of brain tissues were weighed, added to pbs buffer solution, added with appropriate amount of protease inhibitor at 4 °C, and stirred well using a homogenizer [31]. Afterwards, the supernatant was centrifuged and evaluated according to the kit's requirements, and all steps were performed according to the kit instructions (INS-20188 Mouse IL-6 ELISA test kit, INS-20174 Mouse IL-1 β ELISA test kit, INS-20852 Mouse TNF- α ELISA test kit). Follow the steps to configure a series of concentration gradients of the standard working solution, measure the standard curve, and measure the concentration of the corresponding samples through the machine. The analysis was performed by GraphPad Prism 9.5 software.

2.9. Measurement of GSH and MDA levels

The frozen tissue was taken and ground well, appropriate amount of buffer and protease inhibitor was added, centrifuged and the supernatant was collected. Determination of superoxide dismutase (SOD), GSH and malondialdehyde (MDA) in the supernatant according to the kit instructions, Assessment of lipid peroxidase levels in mouse brain tissue was used to reflect the occurrence of ferroptosis in mice. The kits are from INSELISA (Hu Bei, INSELISA, catalog: INS-20174, INS-20188, INS-

20852).

2.10. Statistical analysis

All animal groups were coded, and the analyses were performed by two researchers who were blind to the experimental conditions. The results of the experiment satisfied the principle of random sampling, and the data were analyzed and processed by GraphPad Prism (GraphPad Software, Inc., La Jolla, CA, USA). Group comparisons were analyzed using various statistical tests such as t-tests, ANOVA, and Mann-Whitney U test. P values of <0.05 were considered significant.

3. Results

3.1. MAT inhibits disease progression, CNS immune cell infiltration and demyelination

To enhance comprehension of the involvement of OS/ferroptosis in CNS autoimmunity and the impact of MAT treatment on it, we first induced EAE in C57BL/6 mice through the administration of MOG35-55 peptide. As shown in Fig. 1a, the mean clinical score of the EAE mice treated with MAT was significantly reduced when compared to saline-treated EAE control group (p < 0.05). Mice were sacrificed at day 25 p.i. after extensive perfusion, and H&E and LFB staining were performed on the enlarged lumbar spinal cord. Foci of inflammatory infiltration

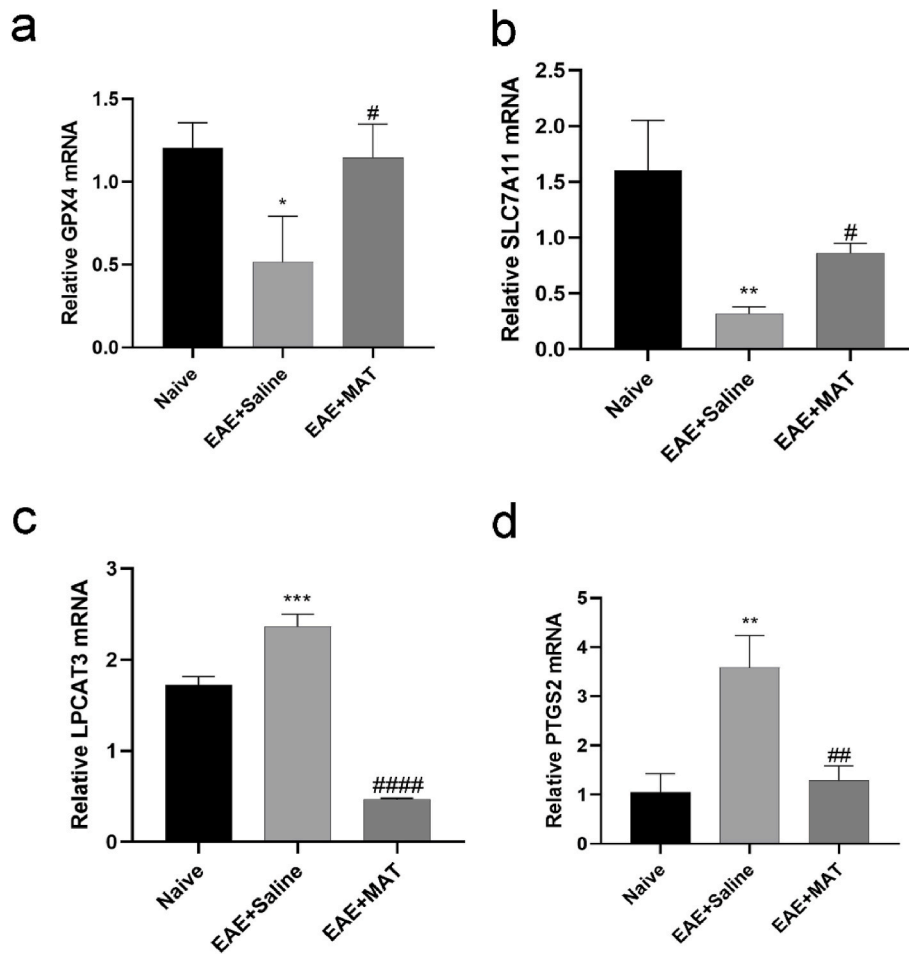


Fig. 3. MAT treatment regulated ferroptosis-related genes. At day 25 p.i., spinal cords from each individual mouse were harvested following extensive perfusion. mRNA expression levels of (a) GPX4, (b) SLC7A11, (c) LPCAT3 and (d) PTGS2 in spinal cord tissues were determined by RT-PCR. Results are represented as mean $\pm$ SD ($n = 3$ per group). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, compared to naive mice; # $P < 0.05$, ## $P < 0.01$, #### $P < 0.0001$, compared to EAE + Saline mice.

and demyelination were observed in EAE mice compared to naïve mice, and MAT treatment significantly reduced the scores of these pathological characteristics of EAE (Fig. 1b–d).

3.2. MAT reduces levels of lipid peroxidation and proinflammatory cytokines in the CNS of EAE mice

We then studied the levels of lipid peroxidation-related molecules in the CNS. The levels of GSH, an essential non-enzymatic antioxidant, and SOD, an antioxidant enzyme in brain tissues of EAE + Saline control mice were largely decreased when compared to the naïve mice, and their expression levels were significantly upregulated by MAT treatment (Fig. 2a and b). In contrast, the level of MDA, an important lipid peroxidation-related molecule, was higher than that in Naïve group, and its expression was significantly inhibited by MAT treatment (Fig. 2c). Consistent with the level of MDA, CNS of saline-treated mice contained increased levels of proinflammatory cytokines IL-6, TNF- α and IL-1 β , which were significantly reduced upon MAT treatment (Fig. 2d–f). These data indicate that MAT treatment reduced the expression of lipid peroxidation, together with reduced levels of proinflammatory cytokines, in the CNS of EAE mice.

3.3. MAT treatment regulates the ferroptosis-related genes

In order to provide additional evidence supporting the assertion that MAT diminishes ferroptosis in the spinal cord of EAE mice through modulation of the SLC7A11-GPX4 axis, we assessed the mRNA

expression levels of GPX4 and SLC7A11. The findings of our study indicate that MAT effectively reversed the downregulation of genes, including GPX4 and SLC7A11, in EAE mice ($p < 0.01$, Fig. 3a and b). Additionally, MAT was found to counteract the upregulation of genes such as LPCAT3 and PTGS2 in EAE mice, thereby inhibiting the arachidonic acid peroxidation pathway ($p < 0.01$, Fig. 3c and d). These results collectively support the conclusion that MAT has the potential to alleviate EAE symptoms by restoring the SLC7A11-GPX4 signaling axis.

3.4. MAT treatment alters expression of ferroptosis proteins

Subsequently, we conducted Western blot analysis to determine the protein expression levels of ferritin, LPCAT3, PTGS2, SLC7A11, and GPX4, which are markers associated with lipid peroxidation and ferroptosis [32], in mice with EAE. As anticipated, the expression of ferritin protein was decreased, whereas the expression levels of LPCAT3 and PTGS2 proteins were increased in EAE mice (Fig. 4a–f). Finally, MAT was employed as a treatment modality for mice. The results demonstrate that MAT effectively counteracted the reduction of Ferritin and elevation of LPCAT3 and PTGS2 in EAE mice (Fig. 4c–f), suggesting a potential pivotal role of Ferritin in EAE pathogenesis and the regulatory capability of MAT on the expression levels of these proteins.

3.5. MAT treatment reduces lipid peroxidation and inflammation of microglia

Given the crucial role of microglia in the pathogenesis and

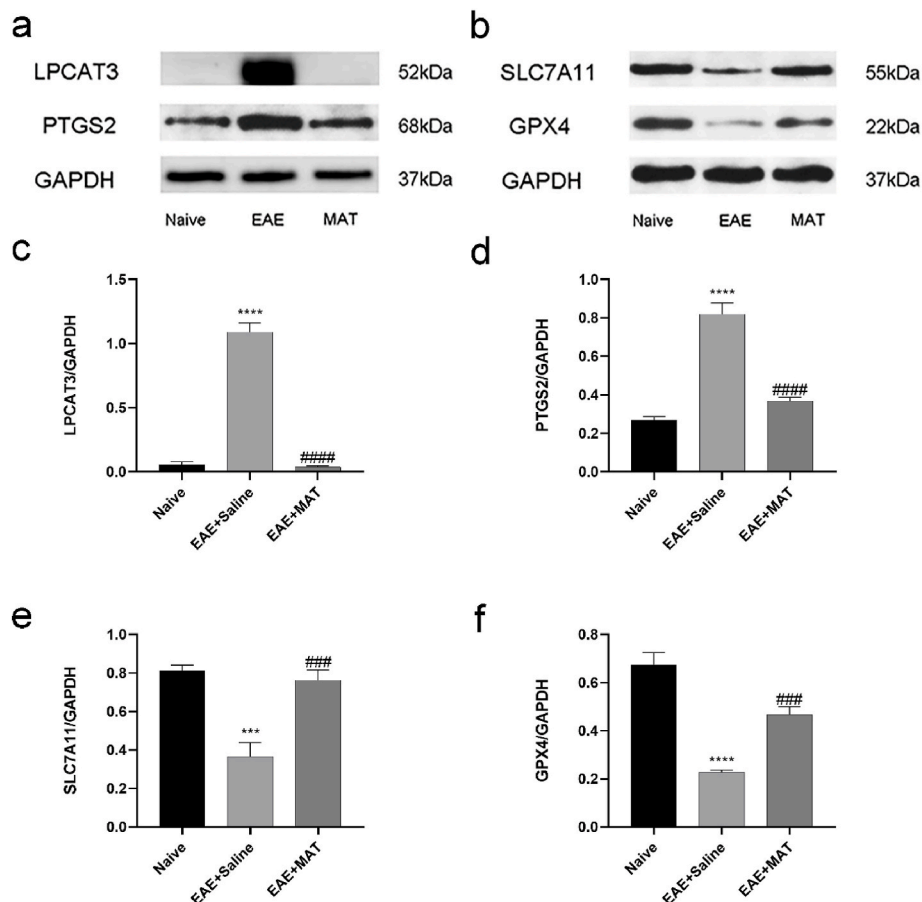


Fig. 4. MAT treatment altered expression of ferroptosis proteins. At day 25 p.i., spinal cords from each individual mouse were harvested following extensive perfusion. (a, b) Levels of ferroptosis-related markers in the spinal cord of mice were further validated with their protein expression by Western blot. Ratios of gray value of LPCAT3 (c), PTGS2 (d), SLC7A11 (e) and GPX4 (f) to GAPDH were calculated. Data are shown as mean $\pm$ SD ($n = 3$ per group). *** $P < 0.001$, **** $P < 0.0001$, compared to naive mice; ### $P < 0.001$, #### $P < 0.0001$, compared to EAE + Saline mice.

modulation of MS/EAE [1], and ferroptosis in the phenotype and function of microglia [14,33], we then determined the extent of microglia expression of lipoxygenases (LOX) and prostaglandin-endoperoxide synthase (PTGS), two key enzymes that regulate the oxidation of unsaturated fatty acids to oxylipins that play a crucial role in the lipid peroxidation process during ferroptosis [34,35]. By immunofluorescence double staining (Fig. 5a and b), increased levels of LOX and PTGS expression were revealed in IBA-1+ microglia of saline-treated EAE mice compared to those of naive mice, and MAT treatment effectively downregulated the numbers of these cells (Fig. 5c and d).

To confirm the relationship of ferroptosis-related molecules with proinflammatory phenotype of microglia in EAE, and the impact of MAT treatment on these mediators, we used immunofluorescence to detect the expression of IL-6 (Fig. 6a) and TNF- α (Fig. 6b), two crucial proinflammatory cytokines for MS/EAE [1,36], in microglial cells. A large increase in the expression of IL-6 and TNF- α in microglia from the Saline-treated group, and these changes were significantly reversed after MAT treatment (Fig. 6a–d).

4. Discussion

In the present study, we provide evidence that, while the CNS of EAE mice contained increased expression of OS/ferroptosis-related molecules but reduced levels of those inhibitory molecules, this was significantly reversed by MAT treatment. Further, MAT treatment significantly reduced the levels of lipid peroxidation and ferroptosis in microglia, together with reduced proinflammatory cytokines. These results,

together with our previous observation that administrating Fer-1, an inhibitor of ferritin phagocytosis, effectively alleviated clinical severity of EAE and decreased lipid peroxidation of oligodendrocytes [11], demonstrate a pathogenic role of ferroptosis in CNS inflammatory demyelination. Further, our present study provide evidence for a link of ferroptosis with proinflammatory phenotype of microglia, and regulating these imbalanced processes confers an important mechanism underlying the therapeutic effect of MAT on EAE.

It has been found that infiltration of inflammatory cells enhances the synthesis of reactive oxygen species (ROS) in targeted tissues [37–39], which leads to imbalanced redox processes and impaired GSH synthesis. In the inflammatory condition, antioxidant system undergoes transport disorders, leading to a decrease in the ability to transport cysteine, and this results in a shortage of cysteine, the raw material for intracellular synthesis of glutathione [40,41]. As a result, the affected cells are unable to synthesize enough glutathione to alleviate the accumulation of intracellular ROS caused by peroxides. Consistent with these results, we have in the present study observed elevated levels of inflammatory mediators (e.g., IL-6, TNF- α and IL-1 β) and lipid peroxidation/ferroptosis-related molecules, e.g., MDA, LPCAT3, PTGS2 and LOX in the CNS of EAE mice, which are important for demyelination and CNS damage. The significant reduction of their expression levels in the CNS by MAT treatment would, thus, greatly alleviating oxidative stress in neural cells, e.g., oligodendrocytes and microglia, thus protecting CNS from demyelination and further axonal damage.

In living organisms, SOD, GSH, and GPX4 function as endogenous antioxidant factors to maintain the redox balance within the cellular environment [42,43]. Glutathione serves as a substrate for GPX4; once

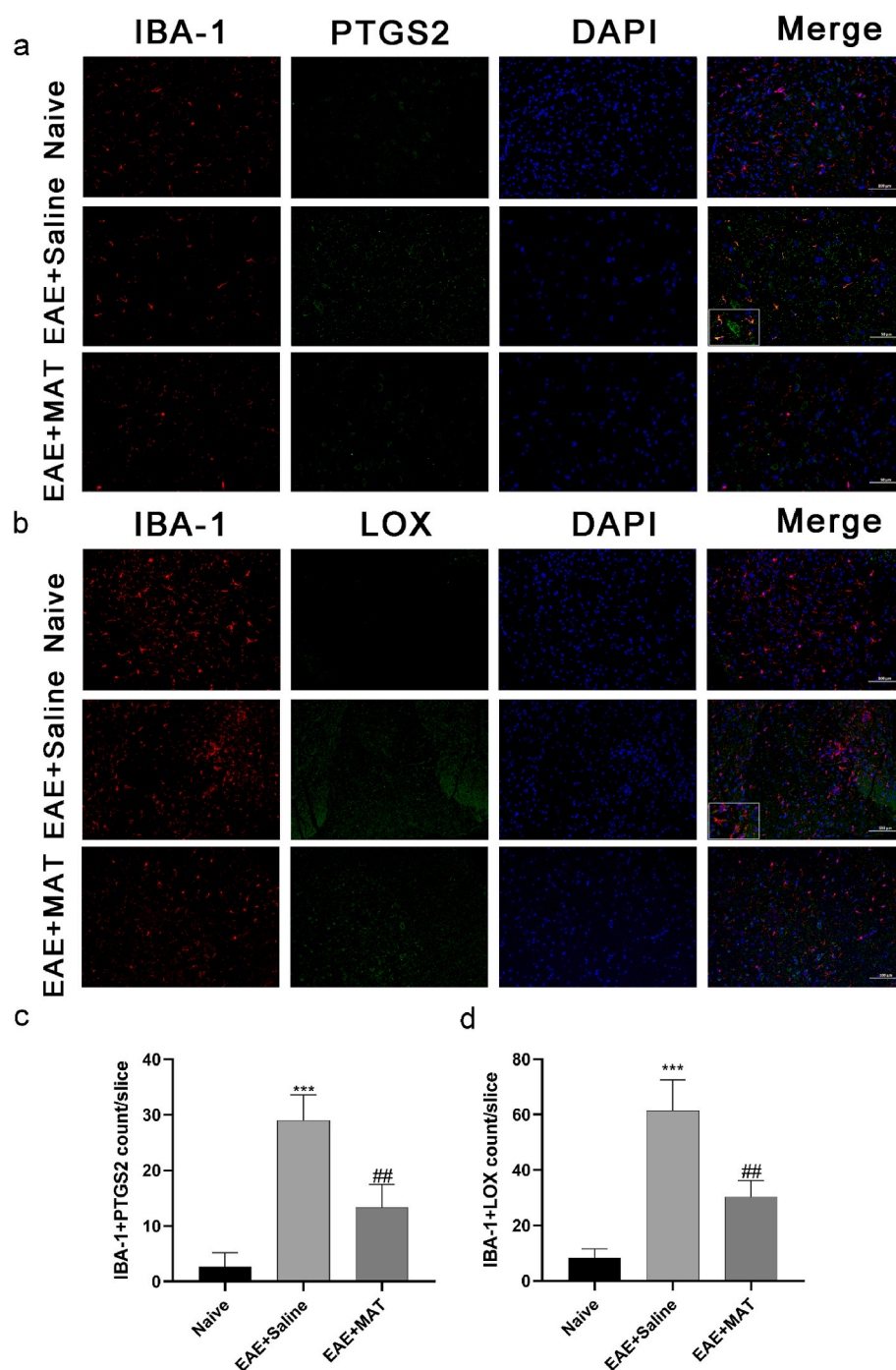


Fig. 5. PTGS2 and LOX expression in microglia. At day 25 p.i., spinal cords from each individual mouse were harvested following extensive perfusion. Expression of PTGS2 (a; green) and LOX (b, green) on microglia (IBA-1, red) was determined by immunofluorescence double staining ($\times 200$). Quantitative analyses for IBA-1⁺PTGS2⁺ (c) and IBA-1⁺LOX⁺ (d) cells were analyzed. Data are shown as mean $\pm$ SD ($n = 3$ per group). *** $P < 0.001$, compared to naive mice; ## $P < 0.01$, compared to EAE + Saline-treated mice. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

depleted, the activity of GPX4 decreases, leading to an increase in lipid peroxidation reactions and the risk of ferroptosis and inflammation [41]. SLC7A11, which is highly expressed in the CNS and the immune system, exchanges glutamate and cystine in a 1:1 ratio and supplies enough substrate for intracellular GSH synthesis [16]. This process contributes to the inhibition of oxidative damage caused by oxidative radicals and reduces cellular OS and ferroptosis. We found that the protein expression of these molecules was significantly reduced in the CNS of EAE mice. As a result, the scavenging capacity for intracellularly

generated ROS was diminished, with substantially elevated levels of ferroptosis-related proteins, e.g., LPCAT3 and PTGS2. These factors make the cells more susceptible to lipid peroxidation. Therefore, upregulation of reduced levels of these endogenous antioxidant factors by MAT treatment will overturn this pathogenic process, leading to reduced CNS inflammation and OS/ferroptosis-induced tissue damage, probably being a novel mechanism of action of this herbal medication.

Microglia, as resident macrophages of the CNS, play an important role in immune surveillance and repair mechanisms following

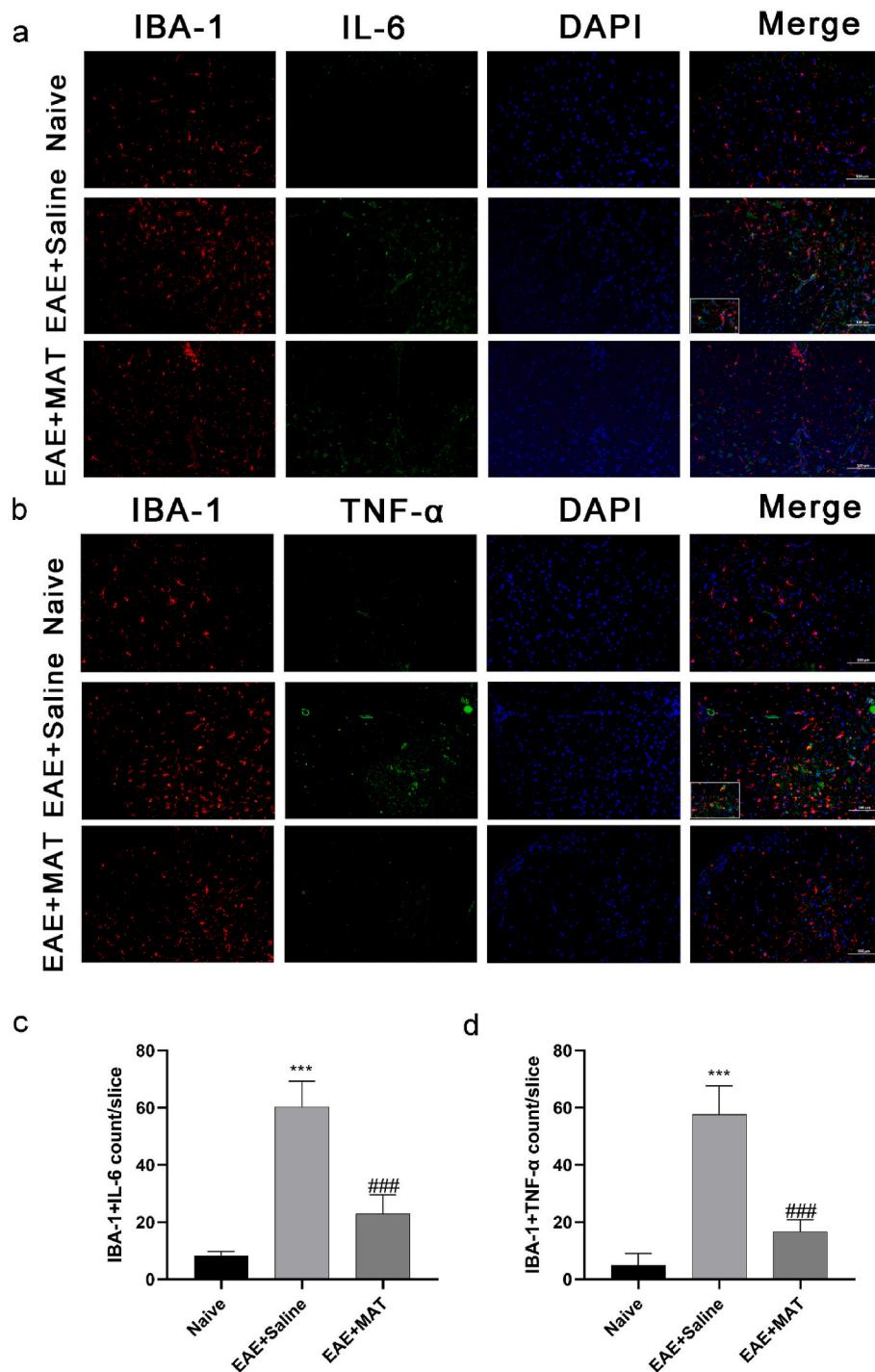


Fig. 6. Expression of pro-inflammatory cytokines in microglia. At day 25 p.i., spinal cords from each individual mouse were harvested following extensive perfusion. Expression (a) IL-6 (green) and (b) TNF-α (green) on microglia (IBA-1, red) were determined by immunofluorescence double staining (× 200). Quantitative analyses for IBA-1⁺IL-6⁺ (c) and IBA-1⁺TNF-α⁺ (d) cells. Data are shown as mean ± SD (n = 3 per group). ****P* < 0.001, compared to naive mice; ###*P* < 0.001, compared to EAE + Saline-treated mice. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

inflammation-related tissue damage during CNS diseases [44–46]. Microglia can be classified into two phenotypes: those with the pro-inflammatory phenotype are crucial effector cells by secreting large amounts of pro-inflammatory factors that lead to demyelination and axonal damage; in contrast, the anti-inflammatory phenotype, which can secrete immunomodulatory cytokines, promotes oligodendrocyte differentiation and accelerates myelin regeneration [47]. A critical role for microglia iron-overload and ferroptosis has been recently shown in

neurodegeneration [14], and cell iron accumulation causes the peroxidation of cell membrane phospholipids and damages the cell membrane, which leads to cell ferroptosis. This process in microglia results in these cells to release a large number of inflammatory factors [33]. It is noteworthy that IL-6, PTGS2 and lipoxygenase (LOX) have been considered as downstream markers of ferroptosis because they contribute to the cellular pool of lipid hydroperoxides that initiate ferroptosis [48,49]. Consistent with these observations, we in the present study found

increased levels of PTGS2, LOX and IL-6, together with proinflammatory cytokine IL-1 β , in microglia of EAE mice, and these factors were inhibited following MAT treatment. These results support the notion that microglia iron overload and ferroptosis play an important role in neurodegeneration and neuroinflammation, and microglia ferroptosis may be a new target for the prevention and treatment of neuroinflammation [14,33].

In summary, our results indicate that ferroptosis and inflammation could be important mechanisms in CNS autoimmunity, and the effects of MAT to upregulate antioxidant pathway molecules but inhibit ferroptosis-induced microglial activation and inflammation suggests a novel treatment mechanism for CNS inflammatory demyelination and probably other neurodegenerative diseases.

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Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Ethical statement

The animal experimental protocol was conducted according to the guidelines of the Ethics Committee of Zhengzhou University (2022-KY-1211-002) and our reporting follows the recommendations in the ARRIVE guidelines.

CRediT authorship contribution statement

Furui Feng: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xinyu Li:** Methodology, Investigation. **Wenbin Wang:** Writing – review & editing. **Mengmeng Dou:** Writing – review & editing. **Silu Li:** Investigation, Data curation. **Xin Jin:** Validation, Methodology. **Yaojuan Chu:** Writing – review & editing, Funding acquisition. **Lin Zhu:** Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: LinZhu reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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